

Tracing Aging- and Disease-Related Epigenetic Changes Using Segmented DNA Methylation

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PROBLEM & GOAL

- Bulk and single-cell DNA methylation data is extremely sparse and noisy
- Noisiness degrades performance of models (e.g., for age prediction) based on DNA methylation data
- Neighbouring CpGs tend to share methylation state

Hypothesis

Grouping CpGs into clusters will reduce noise and improve methylation-based models

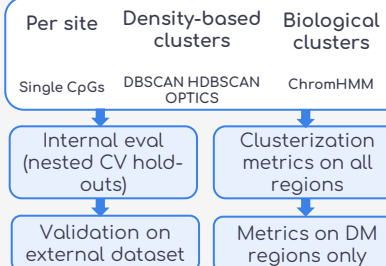
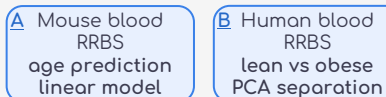
single CpGs



clusterization-based



EXPERIMENTAL DESIGN

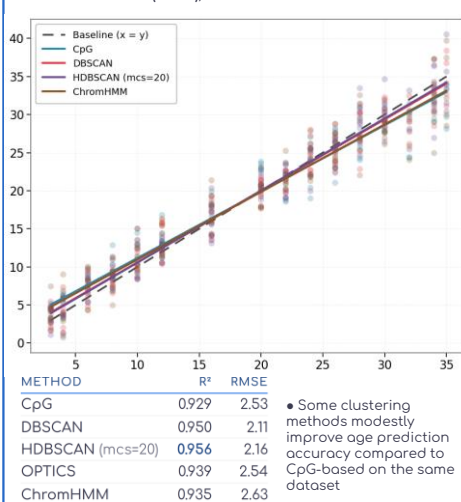


A. SUPERVISED — MOUSE BLOOD AGE CLOCKS

X-axis is always the real age (in months), Y-axis is the predicted age (in months)

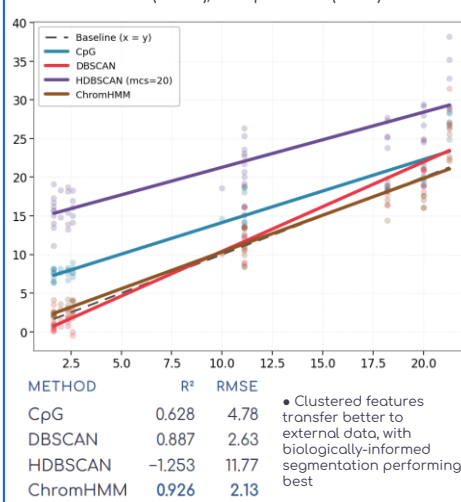
A 1. INTERNAL CROSS-VALIDATION

Petkovich – train (N=132), Petkovich nested 5-fold CV – test



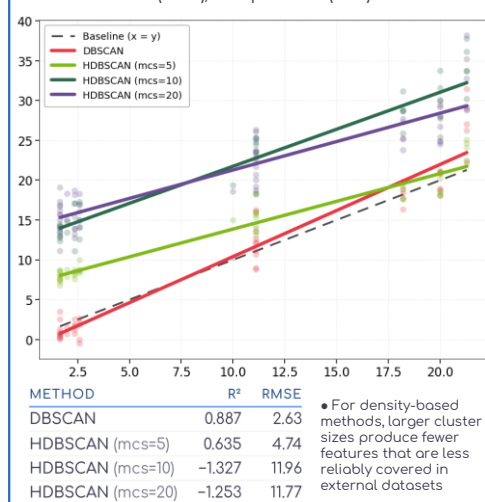
A 2. EXTERNAL VALIDATION

Petkovich – train (N = 132), Thompson – test (N = 50)



A 3. CLUSTER SIZE MATTERS

Petkovich – train (N=132), Thompson – test (N=50)



B. UNSUPERVISED — LEAN VS OBESE SEPARATION

B 1. CLUSTER QUALITY METRICS

DM – differentially methylated regions only; N obese = 8, N lean = 8

FEATURE SET	SILHOUETTE ↑ ALL CPGS	DAVIES-BOULDIN ↓ ALL CPGS	SILHOUETTE ↑ DM CPGS	DAVIES-BOULDIN ↓ DM CPGS
Raw CpG	-0.02	4.5	0.100	2.18
DBSCAN	-0.01	4.2	0.126	1.98
HDBSCAN	-0.02	4.0	0.108	2.10
ChromHMM	-0.01	4.3	0.115	2.05

1. Segmentation produces less noisy, and better-transferable aging clocks compared to site-level CpG models. Density-based methods work but are critically sensitive to hyperparameter choice.
2. Segmentation does not improve group separation in an unsupervised setting. Further validation on an independent dataset is needed, as blood methylation may carry limited signal for obesity.

A 4. BIN-BASED APPROACH

Petkovich – train (N=132), Thompson – test (N=50)

